

AAKASH PORWAL

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Senior Software Engineer | Distributed Systems · Real-Time Data Platforms · Backend Architecture

Backend engineer with 4.5+ years designing and operating high-scale, event-driven microservices and real-time data platforms (300M+ events/day). Strengths in distributed systems, large-scale cost optimization, data engineering, and cross-functional technical leadership.

EXPERIENCE

Tata Digital Private Limited — Bangalore

Dec 2021 – Present

Senior Backend + Data Engineer

Jun 2024 – Present

- **Owned the platform-wide BigQuery cost-optimization strategy end-to-end** — partition-expiry TTLs, dataset lifecycle management, scheduled-job governance, query refactoring, and purging of orphaned/historical datasets (~100 TB managed) — **cutting BigQuery spend ~90%** and establishing cost-governance guardrails; mentored 2 engineers across the initiative.
- **Led an org-wide JSON-to-Avro migration for Kafka event pipelines (~1.5B events)** with Confluent Schema Registry — **cutting storage footprint ~50%** and significantly reducing network bandwidth, while enforcing producer-consumer schema contracts and enabling safe schema evolution.
- **Achieved 100% data parity between Databricks (DWH) and BigQuery** through reconciliation pipelines, automated validation jobs, and continuous monitoring, re-establishing the warehouse as the trusted single source of truth.
- Designed and built both **streaming and micro-batch ingestion architectures on Databricks** (consuming Kafka), with layered **Bronze/Silver/Gold (medallion) PySpark pipelines** powering downstream analytics and ML workloads.
- Built a **resilient backfilling and recovery framework** that automatically restores production data pipelines after Kafka consumer lag, offset resets, upstream outages, and BigQuery ↔ Databricks drift — **cutting recovery from days to hours** while keeping data-consistency SLAs intact during incidents.

Backend Engineer

Dec 2021 – Jun 2024

- **Architected JARVIS, the company's first real-time platform to stitch native-app and WebView journeys into a single unified customer journey** — powering true first-party journey analytics across the Tata Digital ecosystem (Tata Neu, 1mg, BigBasket, Croma, Tata CLiQ, and other Tata brand web journeys). Built on Node.js, TypeScript, and Express over Azure Kubernetes Service; **300M+ events/day** in production.
- Engineered a **multi-sink event-distribution layer** — streaming to BigQuery for product analytics (Mixpanel) and internal BI dashboards, fanning the same events out to Meta, Firebase, Branch, and MoEngage for campaign attribution, audience building, and push notifications, and publishing to Kafka for real-time downstream consumers.
- **Cut BigQuery costs ~70%** in the platform's first optimization phase through table partitioning, clustering, materialized views, data-expiration policies, and migration to the BigQuery Storage API.
- **Owned BigQuery observability end-to-end** — monitoring and proactive alerting across slot utilization, query cost, job health, and failure rates.
- **Migrated event ingestion from Azure EventHub to Kafka** — adding exponential-backoff retries, replication, and a dead-letter (bad-records) topic for durability — **cutting event-processing cost 7x**; mentored 2 interns through the JARVIS build.

Kantar

Jul 2021 – Dec 2021

Senior Analyst — Data Strategy & Engineering

- Modeled data for large-scale Marketing-ROI (MROI) programs across South-East Asia for a **\$150B+ global CPG brand**, delivering competitor analysis and marketing-effectiveness recommendations through advanced analytics.

RoboMQ

Jan 2021 – Jun 2021

Software Engineer Intern

- Built **Spring Boot microservices** in a distributed backend and automated real-time Jira ticket creation from PagerDuty incident triggers via API integrations; developed CRM/ERP/HRM integration flows on an in-house iPaaS.

EDUCATION

B.E., Computer Science & Engineering — Birla Institute of Technology, Mesra

GPA 7.27 · 2021

SKILLS

Languages: TypeScript, JavaScript (Node.js), Python, PySpark, SQL, Java

Backend & Distributed Systems: Microservices, Event-Driven Architecture, REST APIs, Apache Kafka, Avro, Confluent Schema Registry, fault tolerance, high availability, schema governance, Redis, MongoDB

Data Engineering: BigQuery, Databricks, Apache Spark, Medallion (Bronze/Silver/Gold) pipelines, streaming & micro-batch ingestion, data warehousing, reconciliation & backfilling, cost optimization

Cloud, DevOps & Observability: GCP, Azure (AKS, EventHub, API Management, Key Vault), Kubernetes, Docker, NGINX, CI/CD, Git, Prometheus, Grafana